# Skill Relatedness and Chinese Battery Investments in Hungary: An Exploratory Capability-Fit Assessment

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## 1. Research problem, gap and question

Regional diversification is path-dependent: new activities often draw on existing skills and capabilities (Hidalgo et al., 2007). Inter-industry worker flows can reveal a network of skill-related industries (Neffke, Otto & Weyh, 2017), while workers' industry- and location-specific experience can support the survival of pioneer firms (Jara-Figueroa et al., 2018). Yet nonlocal subsidiaries may introduce activities unrelated to local capabilities by using parent-firm resources (Neffke et al., 2018). Evidence from Hungary similarly links foreign-owned firms to regional structural change (Elekes et al., 2019).

These studies provide a framework for assessing recent Chinese battery-supply-chain investments, but the fit of our selected projects with existing manufacturing capabilities remains to be examined. We do not assume that announced investments have already caused regional upgrading.

**Research question.** How do the industries entered by selected Chinese battery-supply-chain investors differ in their positions within a skill-relatedness network, and what broad industrial contexts characterize their Hungarian host counties?

**Hypothesis.** Battery-cell, cathode-material and structural-component activities will differ in network relatedness because they use different technical capabilities. Locally transferable skills may ease entry; where such capabilities are scarce, subsidiaries may instead draw on parent-firm resources. The latter mechanism is proposed, not directly tested.

## 2. Cases and data availability

We focus on five publicly announced Chinese investments (2020–2023), selected to cover battery cells, materials and components across four host counties. BYD vehicle assembly is excluded because it represents a different final manufacturing activity. The announcement year does not imply that production or regional upgrading has occurred.

| Firm | Host county | Activity | Year |
|---|---|---|---|
| CATL | Hajdú-Bihar | Battery cells | 2022 |
| EVE Energy | Hajdú-Bihar | Battery cells | 2023 |
| Sunwoda | Szabolcs-Szatmár-Bereg | Battery cells / systems | 2023 |
| Huayou Cobalt | Komárom-Esztergom | Cathode materials | 2023 |
| Kedali | Pest | Battery structural components | 2020 |

Sources to verify: HIPA/company investment announcements. NACE assignments remain provisional (battery cells: 27.20; Huayou/Kedali: exact activity codes to be confirmed).

The proposed industry network uses a published German worker-flow relatedness matrix (WZ 2008/NACE-based); it is a cross-country proxy, not Hungarian mobility data. KSH county statistics provide broad industry employment shares, but detailed county-by-industry employment needed for an actual regional capability-match index has not been verified as publicly accessible. Investment activities and industry names also require a checked classification crosswalk.

## 3. Network and method

Network nodes are manufacturing industries; links represent above-expected inter-industry worker mobility. Following Neffke et al. (2017, 2018), relatedness is based on observed flows divided by expected flows (controlling for overall flows into and out of each industry). We transform and symmetrise these links for a descriptive network. For each investment industry, we report its normalized degree: the number of positively related manufacturing industries divided by the number of other manufacturing industries. We compare these scores with the manufacturing distribution and test alternative link thresholds.

This score describes an industry's position in the German relatedness network; it is NOT a Hungarian regional capability-match measure. Cases sharing an industry classification necessarily receive the same industry-network score. We separately compare host counties using broad KSH industrial employment data, without inferring county-level skill fit or causal investment effects. If detailed county-by-industry employment becomes available, a later study could estimate related-employment location quotients and regional capability match as in Neffke et al. (2018).

**Plan B.** If industry-code matching or county data remain inadequate, we will construct an ESCO occupation-skill overlap network for selected battery-production jobs (occupations as nodes; shared skills as links). This narrower analysis addresses potential transferable skills, not observed Hungarian worker mobility or regional upgrading.

## 4. Expected results and significance

We expect battery-supply-chain stages to differ in skill-network connections, although project-level rankings are conditional on verified industry codes. The findings could help Hungarian regional development agencies, investment-promotion bodies and vocational training providers identify potential sources of transferable skills and training needs. Given the small, non-random sample and proxy network, the exercise is exploratory and cannot establish effects on regional diversification or actual parent-company knowledge transfer.

## References

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